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RESEARCH ARTICLE

A Weighted Association Rule Mining Method for Predicting HCV-Human Protein Interactions

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Abstract: ***Background****:* Hepatitis C Virus causes the most severe form of chronic liver disease and nearly 200 million people worldwide are estimated to be infected with this virus. Much about the HCV pathogenesis process is still unknown. The study of interactions between HCV and human proteins will lead to deeper understanding of HCV mechanism.

Objective*:* The objective of this paper is to predict potentially new HCV-Human protein interactions using a weighted association rule mining technique.

Methods*:* A new computational method was developed for mining associations within a bipartite graph that was constructed from the HCV-human protein interactions dataset. A new mathematical model was applied to weigh the discovered association rules based on Gene Ontology annotations of viral and human proteins. HCVpro database was used to generate a human-viral bipartite graph that was then analyzed computationally to extract biclusters within the graph. Association rules were extracted from the bipartite graph and weighted using a mathematical model that incorporated information about proteins available from Gene Ontology knowledge base.

Results*:* Forty two new interactions between HCV and human proteins were predicted. Some of these predicted interactions were validated through literature survey and enrichment studies such as Gene ontology-based analysis, pathway- based analysis and disease association based analysis.

Conclusion*:* The methodology developed in this paper can also be used for various other kind of data analysis and hence it carries a wide scope. This will be useful to conduct similar kind of experiments for other disease databases.

Keywords: Weighted association rule mining, gene ontology, HCV-human PPI network, bi-clustering, computational PPI prediction, enrichment studies.

1. INTRODUCTION

Chronic liver disease is a major health problem in many regions. Hepatitis C virus is one of the causative agents of liver diseases. Approximately, 200 million people are chronically infected with HCV and over 0.3 to 0.5 million deaths occur every year due to HCV and its related complications [1]. It is primarily transmitted through blood-blood contact. HCV is an RNA genome containing a single open reading frame which encodes for 3000 amino acids as a polyprotein. This precursor polyprotein is cleaved into eleven viral proteins. At N-terminus, three structural proteins, namely, CORE, E1 and E2 are present. At C-terminus, six nonstructural proteins NS2, NS3, NS4A, NS4B, NS5A, and NS5B are present. In addition to the above structured and non-structured proteins, two more proteins p7 and F have been identified in this genome [2]. These proteins are not only involved in viral replication but also have a role in cellular functions which means that during replication of HCV life cycle, it interferes with the cellular pathways of the host cell and alters the functions of host proteins. These changes are oftenmediated by physical interactions between viral proteins and host proteins.

It is unknown how HCV proteins elicit these changes in the host cell [3]. To address this question, it is essential to study protein interactions between HCV and human proteins. Protein-protein interaction (PPI) data can be collected through in vitro biological experiments such as Yeast two-hybrid systems, mass spectrometry and tandem affinity precipitation. These methods have their own limitations with respect to cost and time. Additionally, the obtained results can be noisy and introduce false positives as well as false negatives [4]. *In silico* approaches can be a more cost-effective alternative and can be widely used to predict new

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protein-protein interactions [5]. Recent computational studies of PPI data primarily focused on single organism such as yeast and humans, while the prediction of PPIs between different organisms has recently emerged [6].

Of particular interest, a recent method of graph bi-clustering combined with association rule mining (ARM) was proposed to address the problem of predicting new protein-protein interactions within a variety of organisms [7, 8]. The method encoded the learned information in the form of association rules. These rules were shown to predict new protein-protein interactions and indirect functional prediction of the proteins. The method showed that it can be potentially useful to help in the identification of drug target [9].

In this paper, HCV-human PPI data was analyzed using a graph bi-clustering algorithm combined with a new association rules mining (ARM) method. A mathematical model was used to weigh the rules based on Gene Ontology (GO) annotations. The proposed method has two main advantages over state-of-the-art methods. First, the method allows for utilization of prior knowledge-based annotations of proteins, and therefore, more informed predictions can be made when applying the association rules. Second, the method helps gaining insight into shared characteristics of proteins (in terms of their GO annotations) when the prediction rules are applied. New predicted HCV-human protein interactions were predicted based on the discovered association rules and some predicted interactions were validated through a literature survey. Further enrichment studies such as Gene ontology-based analysis, pathway-based analysis and disease association based analysis have been performed for validation purpose. This analysis reveals that human proteins interacting with the HCV proteins in the predicted network possess similar biological activities. Thus the proposed weighted association rule mining has the potential to be adapted to incorporate multi-faceted knowledge sources and to be generally applicable to association rules problems, such as drug-protein interactions.

2. MATERIALS AND METHODS

In this section, we describe data collection and preprocessing method together with a computational method to mine association rules in sparse datasets such as HCVPro. The data preprocessing steps aimed to construct an M×N matrix representation of a bipartite graph that has only two types of nodes: human protein nodes and viral protein nodes. In the M×N matrix, rows represented human proteins and columns represented HCV proteins. Then, this matrix and its transpose were used by an algorithm called BiMax to extract biclusters within the matrices.

These biclusters were analyzed by a new probabilistic-based association rule mining algorithm to uncover potentially new associations in the data. The method incorporated information about proteins from Gene Ontology knowledge base to improve scoring (weighting) of the discoveredassociation rules to overcome data sparseness problem with HCVPro database. Information about proteins from Gene Ontology helped improving scores ofthe extractedassociationrules that would have been overlooked (low-scored) if the data were analyzed in its raw form (without external knowledge from Gene Ontology). Fig. **1** represents an overview of the proposed method.

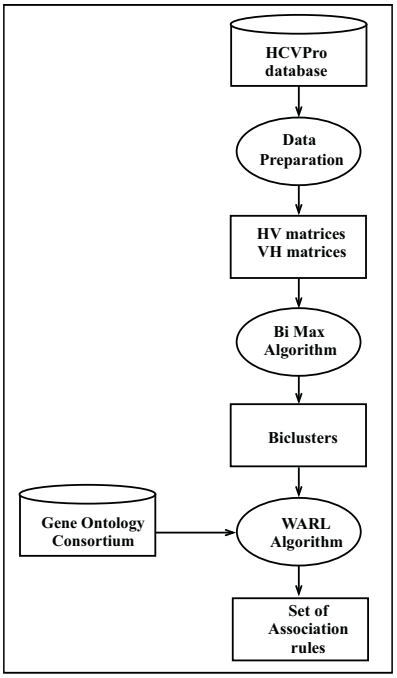


Fig. (1). Pictorial representation of the proposed method.

2.1. Data and Knowledge Resources

In this study, a new association rules for predicting new viral-human protein interactions were created using HCVpro data. HCVPro [10] is a manually curated database created from biomedical literature and protein databases including Virus Mint, VirHostNet and BIND. It contains 11 HCV proteins, 455 human proteins and 549 interactions between HCV proteins and human proteins. After removing redundant interactions, 524 interactions were studied. A bipartite graph was created using these interactions. The graph had two types of nodes: HCV viral proteins and human proteins. An edge between two protein nodes represents an interaction. In this bipartite graph, no edges were allowed between nodes of the same type. Edges did not have weights. Since the total number of nodes in this bipartite graph is 466 and the number of edges is 524, the data set is very sparse.

The HCVpro data was processed in two ways. Two binary incidence matrices were created from HCVpro database namely HV incidence matrix with rows representing human proteins and columns representing viral proteins and the VH matrix which is the transpose of HV matrix. Matrix entries had values 1 or 0 indicating whether the corresponding entries were interacting. This pair of matrices was analyzed using BiMax algorithm to extract biclusters (sub-matrices). These biclusters represented high frequent associations in the given dataset. A new probability-based method was used to generate association rules based on the biclusters and Gene Ontology (GO) annotations of both human and viral proteins. The generated association rules were then combined to predict potentially new viral-human protein interactions.

Due to data sparseness, knowledge-based annotations from Gene Ontology database were used to augment the

biclusters of human-viral proteins in a way that enabled inferring more rules. In its present form, the current dataset does not provide enough support and confidence values for association rule mining algorithm. The idea was to modify the scoring of biclusters to rely more on frequency of molecular function annotation and less on cluster size. This way, the score of a small bicluster that contains some frequent molecular function annotations among its proteins would get more weight and thus would be used effectively to generate association rules.

In this study, in addition to data resources, a knowledge base resource was used to increase the generalization capability of the method to predict new viral-human protein interactions. Annotations from Gene Ontology (GO) consortium [11] were assigned to proteins in HCVpro dataset. Gene Ontology is broadly classified into three components: Molecular function, Biological process and Cellular component. Molecular function describes the biochemical reaction in which a protein is involved in, whereas biological process describes the biological role that a protein plays and cellularcomponent tells about the location of a protein in the cell. In this study, only molecular function annotations were assigned to the proteins in HCVPro.

2.2. Association Rule Mining

Association rule mining (ARM) is a data mining method that extracts descriptive rules in databases [12]. It is typically employed to uncover hidden patterns or regularities among items co-occurring in database records (transactions). ARM can be useful in highlighting some interesting correlations in large volumes of transactions. This method can be divided into two steps. The first step is generating frequent itemsets: finding all sets of items that appear in transactions with frequency greater than a parameter value called minimum support. In the second step, rules are constructed from frequent itemsets. While itemsets need to satisfy the minimum support requirement, a numeric value called confidence score is generated for each rule. More details on association rule mining concepts are given in appendix A.

In bioinformatics research, ARM has been applied in research areas including analysis of gene expression data to relate the expression of one gene associated with the expression of other genes [13] and discovering of the underlying relationships among symptoms, health conditions and diseases [14]. To improve the efficiency of ARM, weighted ARM models have been developed [15]. In this study, we analyzed HCVPro data using a weighted association rule mining method.

2.3. A Two-Step Weighted Association Rule Mining Algorithm

In this section, a new method for weighted association rule mining (WARL) is presented. The method relied on adjacency matrix representation of bipartite graphs where standard frequent item s*et al*gorithms were applied. The problem setting was similar to the work in [16], where bipartite graphs of host-viral protein-protein interactions were represented using notions of transactions and items. For instance, a transaction might be a host protein and items that were observed within the transactions are those viral proteins interacting with this host protein. This problem setting allowed for mining of frequent item sets using popular item set mining algorithms. In the present study, the proposed method had two main steps. The first step was the extraction of closed item sets within binary matrix representations of the bipartite graphs. A closed itemset is a set where none of its proper supersets share its support value [17]. Closed itemsets were utilized in generating association rules with high confidence and low redundancy [18]. In the second step, association rules were weighed using a mathematical model that allows for incorporation of annotations from a knowledge base. This knowledge base provided prior information about transactions when the rules are applied. These prior information enhance the decision making process of predicting future associations instead of relying only on the item set frequencies. Thus, the proposed method is a two dimension model that considers vertical information about items and horizontal information about transactions.

2.3.1. Step 1: Mining of Association Rules Using Graph Biclustering

In the first step, a biclustering algorithm, BiMax, identifies biclusters (within a binary matrix) that satisfy specific criterion such as only finding biclusters with minimum number of rows [19]. This minimum number of rows serves as a value of the support parameter in association rule generation method. Another important feature of BiMax is that it generates closed itemsets, which are more efficient to extract within dense matrices. Given a set of Biclusters, association rules can be generated from each bicluster as follows. Let S denote the column items of a bicluster with $|S| = n$.

$$R = \{A \Rightarrow C : \forall i \in S, S - \{i\}, C = \{i\}\} \quad (1)$$

For each item i in S, the consequent part of the rule is $C=\{i\}$, and antecedent set (A) consists of the remaining elements of S leaving item i out. While closed itemsets are useful in reducing data redundancy, they pose a challenge in the context of mining association rules within bipartite graphs. In some sparse bipartite graphs with very skew distribution of vertex degree, the extracted closed itemsets were observed to have both high cardinality because of too few vertices with high degree and low support score of the rules being generated from such sets.The skew distribution of vertex degree is expected, given that HCV dataset has fewer viral proteins and many human proteins.

To overcome skew degree distributions problem, a post processing method is proposed to systematically expand sparse closed itemsets extracted using BiMax to smaller, more frequent subsets (and hence having higher support values) from which rules can be generated. For each member of the smaller subsets, an association rule is created using the same steps above in Equation (1). Only subsets of closed itemsets that have frequency higher than the superset are considered for further analysis and rule generation. For instance, if $S' \subset S$, then $S'$is considered for further rule generation only if $|S'| < |S|$. Subsets were obtained by generating permutations of the items in the closed item sets. We limited subsets to four elements of size. These subsets, in addition to the closed superset of items were used to generate association rules in the same way as per Equation (1).

2.3.2. Step 2: Weighing of Association Rules Using Gene Ontology (Molecular Function)

One way to apply an association rule of the form $A \Rightarrow C$ is to predict that a transaction T containing items of antecedent set A will also contain items in the consequent set C. This decision process is based on the confidence score of the rule. Within that setting, no prior information about the transactions are used in such predictions. However, there might be some relevant meta data about a given transaction that can be useful to boost the confidence score by utilizing prior information. For instance, in e-commerce contexts, gender, age, marital status, and other information might be associated with the agent (*e.g.* customer) performing the transaction. This meta data/information can be incorporated into a mathematical model that is proposed here. The idea is based on the assumption that there is a hidden dependency of an item set on meta data of transactions. This statistical dependency can be quantified as a probability model that can contribute to the overall confidence score of an association rule. This model will contextualize rules when applied, instead of statically applying the rule without knowing the context.

Let $P(C, T \mid A)$ represent the probability of observing an item set C together with a transaction T, given that another set of items A is observed. Using Bayes' rule and based on the assumption that C and T are conditionally independent on A, this probability can be factored as follows

$$P(C,T \mid A) \propto P(C \mid A) * P(T \mid A) \tag{2}$$

The probability value $P(C|A)$ can be obtained readily from the confidence score of an association rule of the form $A \Rightarrow$C as described in step one of the method. In other words, P(C|A) = Conf(A=>C) (see appendix A for details on calculating support and confidence scores of association rules). The value $P(T|A)$ is the probability of observing a transaction T given a set of items A. The probability $P(T|A)$ does not represent a meaningful quantity by itself if T is just a transaction identifier. A more interesting probability model would be one that represents how much the meta data of a transaction T depends on an item set A. Let $M(T)$ be a function that retrieves (or synthesizes) meta data about a transaction T.

$$M(T) = t_1, t_2, \cdots t_n \tag{3}$$

In this paper, the set of features used are Gene Ontology (GO) molecular function annotations associated with a protein representing the transaction for which a prediction is to be made. Specifically,

$$\begin{aligned} &P(C,T \mid A) \propto P(C \mid A) * P(T \mid A) \\ &= P(C \mid A) * P(go_1, go_2, \cdots go_n \mid A) \end{aligned} \tag{4}$$

Where go_1, go_2, .go_n are molecular functions of protein T.

Thus, the GO annotations allow for defining a context in which the rule is applied to predict whether proteins C and T are interacting. If a protein representing a transaction T is shown to share more molecular functions with the group of transactions (protein) that supports the rule, then the likelihood of applying the rule would increase from a model standpoint. More details about the proposed model and the knowledge sources used are described in appendix B.

2.4. Experimental Evaluation

The WARL method was applied to the problem of mining association rules from bipartite graphs of heterogeneous protein-protein interactions. In this model, C and T are viral/host proteins and $P(C,T|A)$ can be used to predict whether a protein C can be interacting with protein T. In the bipartite graph representing host-viral protein-protein interactions, two types of nodes exist: a human protein type and a viral protein type. Links between nodes indicate interactions between pairs of proteins. No links are allowed between same type proteins. In this study, bipartite graphs were represented by binary matrices where rows indicate one protein type and columns represent the other protein type. We have two matrices HV matrix where rows indicate human proteins and columns indicate viral proteins. The transpose of this matrix is VH matrix. In a previous study [16], mining association rules within two matrices representing the same bipartite graph was found to be beneficial as it was observed that there were few overlapping between the rule set extracted using each matrix, and it was thus possible to combine rules from the two matrices. In this study, we followed the same method of using two matrices. Starting with HV and VH matrices, two sets of association rules were created by first applying BiMax algorithm to create Closed Item sets, and then weighed by a mathematical model based on a probability estimated from gene ontology annotations as meta data of transactions (proteins) in the data set.

In this study, Gene Ontology annotations were used as feature set of a transaction $M(T)$. Hence, $M(T)=go_1, go_2. go_n$ represented the Gene Ontology (GO) annotations that are observed with a transaction representing a protein. The probability model $P(M(T)|A)$ was expressed as a function of the probability $P(go|A)$ and this model is described below:

2.4.1. Estimating P(go|A) Model

In HV and VH matrices, proteins that represented transactions were annotated by molecular function annotation from Gene Ontology knowledge base. For each antecedent set A in an association rule, molecular function annotations of transaction's proteins were used to build a probability model $P(go|A)$. Also, for the set of transactions supporting antecedent set A, the maximum number of observed annotations is stored in a separate table. This number would be used then in calculating the penalty factor when the model was applied to test transactions that have too few of GO annotations associated with A.

Estimation of $P(go|A)$ was done as follows. For each rule $A \Rightarrow C$, GO labels of rows consisting of proteins that appeared both with antecedent set A and consequent set C were used to estimate probability model $P(go|A)$. Transactions that appeared with set A and not with set C were treated as test or candidate (predicted) proteins that may interact with proteins in the set C. The later transactions are evaluated using $P(go|A)$ model according to Equation (B6) given in appendix B.

2.4.2. Evaluating Association Rules Using Estimated P(go|A) Model

Each rule in the final set is evaluated as follows. The probability value $P(C,M(T)|A)$ is calculated for transactions

that are covered by both sets A and C (training set), and for transactions that cover A and does not cover C (predicted set). Predictions with $P(C,M(T)|A)$ score higher than average value of $P(C,M(T)|A)$ of training transactions were considered for further evaluation through published evidence and enrichment studies based on Gene Ontology, pathway and disease association.

2.5. Enrichment Analysis of the Predicted Interactions

To investigate the biological characteristics of the human proteins that interact with the viral proteins, enrichment analysis methods such as Gene ontology (using Biological process and cellular component terms), pathway enrichment analysis (using KEGG database) and disease association analysis (using OMIM database) were performed. For this purpose, web based tools like DAVIDv6.7 [20, 21] and REVIGO [22] were used. DAVIDv6.7 (The Database for Annotation, Visualization and Integrated Discovery) is a resource that provides functional annotation tool for investigating the biological meaning behind a large set of genes. It takes as input a list of genes. REVIGO is a web server tool that can be used to remove the redundant GO terms based on a semantic similarity algorithm. It takes as input a list of GO terms with P-values and displays the output with dispensability. Enrichment threshold value of certain p value and gene count > 2 was set to identify the most important GO terms and pathway terms in the study. The results of these experiments are reported in results section.

3. RESULTS

In this section, we have discussed about the results obtained using WARL method.WARL was applied to HV matrix and its transpose, VH matrix, to generate a set of association rules for predicting new potential interactions. An association rule was applied to a candidate input protein whenever the items of antecedent set of the rules have non-zero entries in the incidence matrix. If a protein record matches the antecedent set, the proteins in the consequent set of the rules were predicted to interact with the candidate protein. After that, the mathematical model for rule weighing was applied to confirm whether a prediction should be rejected. Initially, there were 847 candidate predictions within the HV matrix and 88 candidate predictions in VH matrix. Only candidate predictions above an empirical threshold value were not rejected. Several values were tried and those resulting in too many or too few non-rejected predictions were not used. A threshold value of $1E$-14 was finally used to filter out predictions.

3.1. Results on HV and VH Matrix

After screening off predictions based on the threshold value, the resulting predictions were reduced from 847 to 30 from the HV matrix. Table **1** shows the final set of 30 predictions based on HV matrix. From the table it is clearly seen that only 9 HCV proteins are in the consequent part of the rules. Hence according to the 30 predicted rules, out of 455 human proteins, 15 of them are found to interact with the above 9 HCV proteins. To the above 15 Human proteins, NS3 interacts with 8 of them which shows that NS3 has a higher degree in the HV matrix. Thus the new predictions involve 15 Human proteins and 9 HCV proteins. Similarly, out of the 88 predictions in the VH matrix, 46 of them exceed the threshold value. Consequent part of the above 46 rules involve 24 Human proteins which interacts with seven HCV proteins. Table **2** shows the final set of 46 predictions based on VH matrix.

Table 1. Final set of 30 association rules among HCV proteins predicted from HV matrix.

Human-Viral Predicted Pairs	Applied Association Rule
CD209,NS2	{E1}=>{NS2}
CEP250,NS3	{NS5A,NS5B}=>{NS3}
CLEC4M,CORE	{E1,E2}=>{CORE}
CLEC4M,NS2	{E1,E2}=>{NS2}
CLEC4M,NS5B	{E1,E2}=>{NS5B}
CREB3,NS3	{CORE,p7}=>{NS3}
CREB3,NS5B	{NS4A,p7}=>{NS5B}
DDX3X,NS3	{CORE}=>{NS3}
DDX3X,NS5B	{CORE}=>{NS5B}
DDX5,E2	{CORE,NS5B}=>{E2}
DDX5,NS2	{CORE,NS5B}=>{NS2}
DDX5,NS4A	{CORE,NS5B}=>{NS4A}
EIF2AK2,E1	{CORE,E2}=>{E1}
EIF2AK2,NS2	{CORE,E2}=>{NS2}
EIF2AK2,NS3	{CORE,NS5A}=>{NS3}
EIF2AK2,NS4A	{CORE,E2}=>{NS4A}
EIF2AK2,NS5B	{CORE,E2}=>{NS5B}
FMNL1,CORE	{p7}=>{CORE}
FMNL1,NS3	{p7}=>{NS3}
FMNL1,NS4A	{p7}=>{NS4A}
FTH1,CORE	{NS3,NS5A}=>{CORE}
FTH1,NS5B	{NS3,NS5A}=>{NS5B}
HOXD8,NS3	{NS2}=>{NS3}
HOXD8,p7	{NS4A,NS5B}=>{p7}
NR4A1,NS3	{E1}=>{NS3}
NR4A1,p7	{CORE,NS4A}=>{p7}
PTBP2,NS5A	{NS3,NS5B}=>{NS5A}
SETD2,NS3	{CORE,NS2}=>{NS3}
SLIT2,CORE	{NS3,p7}=>{CORE}
TRIM27,CORE	{NS2,NS3}=>{CORE}

Table 2. Final set of 46 association rules among Human proteins predicted from VH matrix.

Viral-Human Predicted Pairs	Association Rule
E1,CREB3	{NR4A1}=>{CREB3}
NS2,CALR	{SETD2}=>{CALR}
NS2,CANX	{NR4A1,SETD2}=>{CANX}
NS2,EIF2AK2	{HOXD8,NR4A1}=>{EIF2AK2}
NS2,EIF2AK2	{SETD2}=>{EIF2AK2}
NS2,HSPA5	{SETD2}=>{HSPA5}
NS3,CEP250	{PSMB9}=>{CEP250}
NS3,CREB3	{FBLN2}=>{CREB3}
NS3,EIF2AK2	{TP53}=>{EIF2AK2}
NS3,FBXL2	{PSMB9}=>{FBXL2}
NS3,HOXD8	{FBLN5}=>{HOXD8}
NS3,JAK1	{TP53}=>{JAK1}
NS3,MOBK1B	{PSMB9}=>{MOBK1B}
NS3,NR4A1	{FBLN5}=>{NR4A1}
NS3,SETD2	{FBLN5}=>{SETD2}
NS3,SHARPIN	{PSMB9}=>{SHARPIN}
NS3,STAT1	{TP53}=>{STAT1}
NS3,TBP	{TP53}=>{TBP}
NS3,TP53BP2	{TP53}=>{TP53BP2}
NS3,TRADD	{TP53}=>{TRADD}
NS3,TRAF2	{TP53}=>{TRAF2}
NS3,VAPA	{PSMB9}=>{VAPA}
NS3,VAPB	{PSMB9}=>{VAPB}
NS4A,EIF2AK2	{HOXD8}=>{EIF2AK2}
NS4A,EIF2AK2	{NR4A1}=>{EIF2AK2}
NS5A,HOXD8	{EIF2AK2}=>{HOXD8}
NS5A,NR4A1	{EIF2AK2}=>{NR4A1}
NS5A,SETD2	{EIF2AK2}=>{SETD2}
NS5B,CALR	{NR4A1,SETD2}=>{CALR}
NS5B,CANX	{NR4A1}=>{CANX}
NS5B,CANX	{SETD2}=>{CANX}
NS5B,CD209	{NR4A1,SETD2}=>{CD209}
NS5B,CLEC4M	{NR4A1,SETD2}=>{CLEC4M}
NS5B,CREB3	{HOXD8,NR4A1}=>{CREB3}
NS5B,CREB3	{UBQLN1}=>{CREB3}
NS5B,EIF2AK2	{HOXD8,SETD2}=>{EIF2AK2}
NS5B,EIF2AK2	{NR4A1,SETD2}=>{EIF2AK2}
NS5B,FBLN5	{HOXD8,NR4A1}=>{FBLN5}
NS5B,FBLN5	{HOXD8,SETD2}=>{FBLN5}
NS5B,FBLN5	{HOXD8}=>{FBLN5}
NS5B,FBLN5	{NR4A1,SETD2}=>{FBLN5}
NS5B,HSPA5	{NR4A1,SETD2}=>{HSPA5}
NS5B,HSPA5	{NR4A1}=>{HSPA5}
NS5B,LTF	{NR4A1,SETD2}=>{LTF}
p7,HOXD8	{CREB3}=>{HOXD8}
p7,HOXD8	{UBQLN1}=>{HOXD8}

3.2. Bipartite Graph of the Predicted Interactions

Out of the 30 predicted interactions in HV matrix and 46 predicted interactions in VH matrix, 12 interactions were common in both matrices. Further, 22 interactions were repeated with different antecedent sets. Eliminating these common and repeated interactions, the number of final predicted interactions was reduced to 42. These 42 interactions involved 30 Human proteins and 9 HCV proteins

Fig. **2** represents the predicted bipartite network involving 30 human proteins and 9 HCV proteins. The figure illustrates that there are predominantlytwo viral hub proteins NS3 and NS5B which interact with 13 and 8 Human proteins respectively. The other HCV proteins of the network are NS2, CORE, NS5A, E1, NS4A, E2 and p7 and their degrees are respectively 6, 5, 4, 2, 2, 1 and 1.

3.3. Support Evidence of Predicted Host-Protein Interactions

To verify the newly predicted interactions, we searched PubMed database for recent reports on protein interactions between HCV proteins and human proteins which are not included in the HCVpro database. In HCVpro, DDX3X interacts with the CORE proteins of HCV, whereas the WARL method predicted that DDX3X may be interacting with NS3 and NS5B. In [23], it has been demonstrated by Immunofluorescence and co-immunoprecipitation assay, that NS3 and NS5B were coimmunoprecipitated with DDX3X and IKK in cells transfected with full length of HCV RNA. This supports the existence of the WARL-predicted interactions of DDX3X with NS3 and NS5B. Also, the WARL method predicted that NR4A1 interacts with NS3. It has been shown in [24] that knocking down of NR4A1 expression decreases the expression of NS3 in Huh cell lines. Likewise, the proposed method predicted the interaction of CD209 with CORE and NS3 viral proteins. In [25], it was shown that HCV core and NS3 proteins inhibited the expression of CD1a, CD1b and CD209 during monocyte differentiation to dendritic cells, while increasing some marker characteristics of macrophages and also PD-L1 expression. The mechanism for this inhibition was associated with TLR2 signaling. While most WARL-predicted interactions in this study do not have evidence in literature to date, the existence of these interactions may be plausible given the functional-based similarity that WARL method took into account. These new predictions may pave way to understand the molecular mechanisms of HCV pathogenesis and may lead to novel therapeutic target against HCV.

3.4. Enrichment Analysis of the Predicted Interactions

In this section we discuss about the enrichment studies that were performed for the human proteins predicted to interact with HCV proteins in the predicted bipartite network. The overall results are given in the supplementary File **S1** and summarized results are given in Tables **3**, **4** and **5**.

3.4.1. Interaction with NS3

NS3 is a non-structural HCV protein containing serine protease and RNA helicase. The protease inactivates cellular

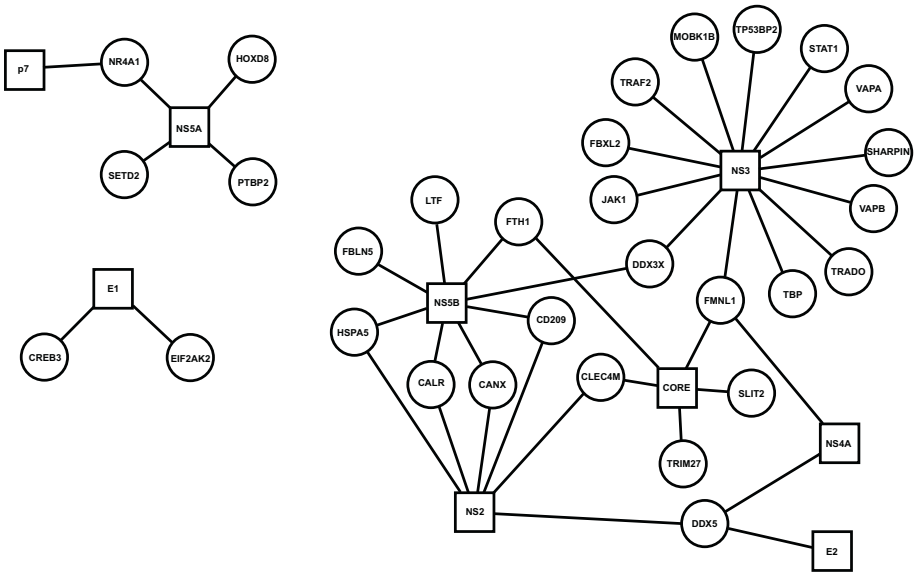


Fig. (2). Bipartite network of predicted HCV-human protein interactions which involves 30 human proteins and 9 HCV proteins generated from HV and VH matrix. Human proteins are represented in circle and HCV proteins are represented in square.

protein required for innate immunity whereas helicase is required for genome replication and virus assembly [26]. From the WARL prediction rules, it was identified that NS3 has a higher degree than other HCV protein and that 13 human proteins are interacting with this protein. To identify the biological relationship between these 13 human proteins, we tested the enrichment analysis using DAVIDv6.7. To remove the redundant terms, the identified GO terms with the P-value were processed using REVIGO. Table **3** shows the significant, non redundant GO terms as well asKEGG pathway studies conducted for the human proteins that are predicted to interact with NS3 of HCV protein with certain threshold value of p and dispensability. It can be shown in table that the groups of human proteins that were predicted to interact with NS3 are related biologically and possess common biological activities. In particular a set of seven human proteins were found to be involved in the biological process of cell death. The proteins involved in this process may participate either as inducers or repressors of the programmed cell death. When NS3 interacts with the above seven human proteins it affects the regular function of the human proteins and results in improper functioning of the immune system. Similarly another group of human proteins are found to participate in the intracellular and cytoplasm of the cellular component terms. These proteins have a role in internal organization of the cell structure. NS3 may interact with these human proteins for their genome assembly and viral replication. Further, it makes them inactive and disrupts their normal activity for their own replication inside the host cells. Apart from gene ontology studies, pathway analysis confirmed that some of the human proteins interacting with NS3 are involved in other pathways like RIG-I-like receptor signaling pathway, Adipocytokine signaling pathway, pathways in cancer and pancreatic cancer. This result suggests that possible viral interaction to human proteins may lead to development of some cancer types. Similarly disease association studies were performed. These proteins are likely to have more chance of being relevant for other diseases such as stomach cancer and tuberculosis. This is important since HCV infection through NS3 may indirectly cause stomach cancer.

3.4.2. Interaction with NS5B

HCV RNA needs NS5B protein for viral replication [26]. Using WARL method, a set of 8 human proteins was found to be interacting with NS5B protein. Thus, NS5B is the second highest degree protein in the predicted interactions. Similar studies based on GO, pathway and disease associations have been done for NS5B and the results are shown in Table **4**. From the table it is noted thatmost of the human proteins were involved in theimmune system process and cellular localization.

When NS5B interacts with these human proteins, they are made inactive and lead to viral spread throughout the body. Further, 87.5 percent proteins are involved in the component of membrane and cytoplasm which is associated with membrane organization process. Interestingly, NS5B is also a membrane-associated protein which obviously depends on these human proteins for viral replication. Hence there is a high chance of interaction of NS5B with these human proteins. From pathway analysis it was found that out of eight human proteins, three of them namely, CALR, CANX and HSPA5 were involved in antigen processing and presentations suggesting that when NS5B interacts with these human proteins, functioning of natural killer cell mediated cytotoxicity fails.

3.4.3. Interaction with NS2

The other hub protein found in the predicted interaction is NS2, cysteine protease [26]. This protein is mainly involved in the processing of NS2-NS3 cleavage and also has some role in virus assembly and release. The six human proteins that were predicted to be interacting with NS2 protein were found to share some common biological activity. The results are given in Table **5**. The table shows that three of these human proteins were located in the

Table 3. Significant and non redundant GO terms and KEGG pathway enrichment analysis of human proteins that are predicted to interact with HCV protein NS3.

GO-ID	Biological Process	No. of Genes	Gene Symbols	P Value < 0.002	Dispensability< 0.1
GO:0008219	Cell death	7	SHARPIN, STAT1, TBP, VAPB TP53BP2, TRADD, TRAF2	6.34E-06	0
GO:0044419	Interspecies interaction between organisms	4	DDX3X,STAT1,TBP,VAPB	0.00116	0
GO:0051704	Multi-organism process	4	DDX3X,STAT1,TBP,VAPB	0.0137	0
GO:0016265	Death	7	SHARPIN, STAT1, TBP, VAPB TP53BP2, TRADD, TRAF2	6.60E-06	0.048
GO:0019221	Cytokine-mediated signaling pathway	3	JAK1,STAT1,TRADD	0.00129	0.078
GO-ID	**Cellular Component**	**No. of Genes**	**Gene Symbols**	**P value < 0.002**	**Dispensability< 0.2**
GO:0005737	Cytoplasm	12	SHARPIN,STAT1,TBP,TP53BP2,TRADD,TRAF2,VAPB,FMNL1,FBXL2,JAK1,VAPA,DDX3X	1.95E-04	0
GO:0005622	Intracellular	12	SHARPIN,STAT1,TBP,TP53BP2,TRADD,TRAF2,VAPB,FMNL1,FBXL2,JAK1,VAPA,DDX3X	0.01716	0.183
KEGG-ID	**Pathway-Based**	**No. of Genes**	**Gene Symbols**	**P Value < 0.1**	
hsa04622	RIG-I-like receptor signaling pathway	3	DDX3X,TRADD,TRAF2	0.00278088	
hsa05200	Pathways in cancer	3	JAK1,STAT1,TRAF2	0.052328966	

Table 4. Significant and non redundant GO terms and KEGG pathway enrichment analysis of human proteins that are predicted to interact with HCV protein NS5B

GO-ID	Biological Process	No. of Genes	Gene Symbols	P Value < 0.02	Dispensability< 0.1
GO:0030005	Cellular di-, tri-valent inorganic cation homeostasis	3	CALR,FTH1,LTF	0.0051	0
GO:0055066	di-, tri-valent inorganic cation homeostasis	3	CALR,FTH1,LTF	0.0056	0
GO:0002376	Immune system process	4	CALR,FTH1,LTF,CD209	0.0099	0
GO-ID	**Cellular Component**	**No. of Genes**	**Gene Symbols**	**P value < 0.05**	**Dispensability< 0.1**
GO:0005576	Extracellular region	4	CALR,FBLN5,LTF,CD209	0.0475	0
GO:0016020	Membrane	7	CALR, FTH1, LTF, CANX, FBLN5, DDX3X, HSPA5	0.0386	0
GO:0005737	Cytoplasm	7	CALR, FTH1, LTF, CANX, FBLN5, DDX3X, HSPA5	0.0401	0.089
KEGG-ID	**Pathway-Based**	**No. of Genes**	**Gene Symbols**	**P Value < 0.002**	
hsa04612	Antigen processing and presentation	3	CANX, CALR, HSPA5	0.00154	

component endoplasmic reticulum. From the Gene ontology consortium (GOC), it was found that NS2 is also located in the same component. Hence there is a high chance of interaction of NS2 with these human proteins.

3.4.4. Interaction with CORE

CORE is a structural protein which has a role in viral capsid formation in Hepatitis C virus [26]. It directly interacts with most cellular proteins and regulates the host cellular functions such as apoptosis and signaling pathway. Using WARL prediction, five human proteins were found to interact with CORE. Similar enrichment analysis has been done and it has been found that all the 5 human proteins are involved in the cellular component organization. As stated earlier, CORE protein has a role in capsid formation.For this process it needs to interact with cellular organization proteins and WARL prediction method shows that all these

Table 5. Significant and non redundant GO terms and KEGG pathway enrichment analysis of human proteins that are predicted to interact with HCV protein NS2.

GO-ID	Biological Process	No. of Genes	Gene Symbols	P Value < 0.05	Dispensability< 0.02
GO:0002376	Immune system process	3	CALR, CLEC4M, CD209	0.0432	0
GO:0048002	Antigen processing and presentation of peptide antigen	3	CALR, CLEC4M, CD209	3.78E-05	0
GO-ID	**Cellular component**	**No. of Genes**	**Gene Symbols**	**P value < 0.05**	**Dispensability< 0.3**
GO:0044425	Membrane part	5	CALR, CLEC4M, CD209, HSPA5, CANX	0.0977	0
GO:0005783	Endoplasmic reticulum	3	CALR, HSPA5, CANX	0.03229	0.279
KEGG-ID	**Pathway-based**	**No. of Genes**	**Gene Symbols**	**P Value < 0.002**	
hsa04612	Antigen processing and presentation	3	CANX, CALR, HSPA5	7.81E-04	

five human proteins are related to cellular organization process. Hence there is a high chance of interaction of these five human proteins with CORE. Also four of these five human proteins are highly associated with component plasma membrane, which plays a role in structural formation.Obviously CORE protein utilizes these protein functions for its viral capsid formation. Interestingly these proteins are also associated with apoptosis pathway which clearly states that there is a high chance of interacting with CORE protein since CORE protein regulates the apoptosis pathway.

3.4.5. Interaction with NS5A

NS5A is a non-structural protein, a component of a replication which is involved in RNA binding [26]. From GOC, it is known that this protein has a role in apoptotic process, interfere in host intracellular interferon activity and regulation of viral transcription. Four human proteins are found to interact with this viral protein. From the GO study of the human proteins, three of them are involved in the regulation of transcription. This biological process is also shared by NS5A protein and hence there is a high chance of interaction between NS5A and the four human proteins. Further these four human proteins and NS5A are also found in Cellular component nucleus, which strengthens our prediction.

3.4.6. Interaction with Other HCV Proteins

The other HCV proteins NS4A, E1, E2 and p7 have a small degree in the predicted network. NS4A is the association factor of NS3 HCV protein which cleaves the HCV proteins. DDX5 and FMNL1 are found to interact with NS4A. The envelop proteins E1 and E2 are involved in virus attachment to host cell and fuse with the host membrane. The human proteins EIF2Ak2 and CREB3 interact with E1 protein in our prediction, whereas DDX5 human protein alone interacts with E2 protein. These human proteins have a role in cell proliferation. If the envelop protein interacts with these proteins then it initiates the virus to multiply within the host cell. Viral protein p7 finds to interact with NR4A1 human protein in the predicted network.

4. DISCUSSION

In this study, the problem of predicting host-viral protein-protein interactions was addressed in the context of association rule mining. Following a methodology described in [16], the data was modeled as bipartite graphs and the BiMaxbiclustering algorithm was used to find closed item sets from which association rules were generated. Due to problems of data sparseness and skewness of the vertex degree distribution of the bipartite graphs, the identified closed itemsets in this study suffered low support values and high set cardinalities. That meant low confidence scores and limited applicability of rules. Because of that many items in test transactions had to be matched against antecedent sets of the rules. To handle this problem, a new method was proposed to expand the closed itemsets found by generating subsets of closed itemsets that are of higher frequency (and hence higher in support and confidence values). These subsets of closed itemsets were used in generating a set of association rules. The generated rules were evaluated in prediction of novel protein protein interactions. A new model was proposed for weighing association rules in a way that benefited from existing knowledge bases such as Gene Ontology. This model put some emphasis on the characteristics of a transaction when applying an association rule. Thus it allowed for considering a context when applying an association rule. This method helped making more informed predictions in a way that can be better than applying the association rules without any context. The proposed method made it possible for predicting new interactions within the dataset.

Expanding closed itemsets by more frequent (and smaller) subsets came in response to the data sparseness problem posed by the dataset subject of the study. Skew degree distributions present a common property of complex networks, where the node degree distribution can be described by power-law. The method proposed in this study can be applied in other contexts where bipartite graph suffers this data sparseness and degree skewness phenomenon.

A key component of the method proposed in this study is the identification of characteristic molecular features (as represented by GO molecular function annotations) of transactions that are associated with antecedents sets $\{A\}$ of

all rules. Using molecular function annotations, the model was aimed to identify what is common among protein sets that support a given association rule. The proposed method relied on the molecular ingredient that is essential to exist for a protein representing a new transaction to interact with the protein identified by the consequent set.

Our method proposed a solution to a problem of a previous method in [16]. That problem was that only new interactions involving human proteins that are present in the dataset could be predicted. Our new model for predicting new protein-protein interaction overcomes this limitation and the method can be applied to proteins transactions outside the database as long as the human proteins have enough highly probable features (GO annotations from the knowledge base) according to $P(M(T)|A)$ model proposed in the methods section. Thus, our method has a more generalization capability by being able to handle future transactions that were not observed during probability estimation. While the proposed model in this study relied on GO annotations of proteinstransactions, it can be shown that other features or combination of features can be used as well. The key is how $P(M(T)|A)$ can be modeled. For instance, $P(M(T)|A)$ can be represented by a combination of features.

$$P(T \mid A) \approx P(M_1(T), M_2(T) \cdots M_m(T) \mid A) = \prod_{i=1}^{m} w_i * P(M_i(T) \mid A) \quad (5)$$

where, w_i represents weight of a feature dimension.

Thus the proposed method has predicted new Viral-Human protein interactions. A number of these new interactions were supported by evidence from recent research publications in this domain. Further enrichment studies such as Gene ontology-based analysis, pathway-based analysis and disease association based analysis were performed for validation purpose. This study reveals that most of the human proteins predicted to interact with the HCV protein share common biological activities like cell death, immune system process and initiation of viral infection. Further, the location of predicted human proteins and viral proteins are also common in some cases. Furthermore some human proteins that are predicted to interact with HCV proteins are related with pathwaysof some type of cancer and suggest that HCV infection may indirectly cause cancer and other related diseases. Although these studies give some biological information of the predicted interaction, still more detailed study is needed. Further experimental studies are needed in this direction to confirm the validity of these predictions for future development.

CONCLUSION

In this paper, we presented a new association rule mining method for sparse datasets about viral-human protein interactions. The method addressed data sparseness issue through using knowledge base to increase the generalization capability of the extracted rules. In this study, Gene Ontology molecular function annotations were used in a mathematical model for weighing rules. The new method allowed for extracting low-support rules from the database and then increasing confidence score of the rule when applied to a protein through prior knowledge represented in Gene Ontology. The findings were evaluated through a survey and few predicted interactions were mentioned in recent research publications. Enrichment analyses also have been performed for validation purpose. The method has the potential to be applied in other areas which include drug-protein interactions and gene-disease associations.

CONSENT FOR PUBLICATION

Not applicable.

CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.

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APPENDIX A

Association Rule Mining in Protein-Protein Interaction Datasets

Let $I = \{i_1, i_2, ..., i_n\}$ be a set whose elements are called items. Let $D = \{t_1, t_2, .t_m\}$ be another set whose elements are called transaction and each transaction t_k is a subset of I. In the context of human-viral protein interaction databases, a transaction database can be defined in a way where viral proteins represent the itemset I, and human proteins represent the transactions that include a selection of members of set I. A protein-protein interaction database can be defined as shown in the example transaction Table **A1**.

Table A1. A hypothetical example of transaction databases representing human-viral protein interactions. Columns represent viral proteins (*V*) and rows represent human proteins (*H*) interacting with viral proteins. A value of 1 indicates interaction, 0 indicates no interaction between respective *H* and *V* proteins.

Transactions (Human Proteins)	V_1	V_2	V_3	V_4
H_1	1	1	0	0
H_2	1	0	1	0
H_3	0	0	0	1
H_4	1	0	1	0
H_5	1	1	1	1

As shown in Table **A1**, each row represents a human protein. The itemset is defined as I = $\{V_1, V_2, V_3, V_4\}$ and the transaction set T = $\{H_1, H_2, H_3, H_4, H_5\}$, where $H_1 = \{V_1, V_2\}, H_2 = \{V_1, V_3\}, H_3 = \{V_4\}, H_4 = \{V_1, V_3\}$ and $H_5 = \{V_1, V_2, V_3, V_4\}$.

Now, an association rule $A \Rightarrow C$ consists of two sets:

1. An antecedent itemset*A* that states a pre-condition to apply the rule.

2. A consequent itemset *C* that are likely to be observed once set *A* is observed.

An association rule has a numeric value called confidence that reflects the degree to which the data suggest that rule. The two parts of the association rule have also a numeric value called support. The support (*A*) is the percentage of rows in the dataset that contains the items of *A*.

For example, as shown in Table **A1**, we have the frequent item set $\{V_1, V_3\}$ with support = 3/5 = 0.6 (*e.g.* V_1 and V_3 were observed in three transactions (H_2, H_4, H_5)).

The confidence of the rule *R* can be computed as:

Conf(*R*) = Support ($A \cup C$)/Support (*A*)

Support ($A \cup C$) is the percentage of rows in the dataset that contain both set *A* and set *C*. The value of Conf(*R*) ranges from 0 to 1.

Back to the above example transaction table, the frequent item set $\{V_1, V_3\}$ can be used to define an association rule $\{V_1\} \Rightarrow \{V_3\}$. The confidence of this rule would be Support ($\{V_1, V_3\}$)/Support ($\{V_1\}$) = 0.6 /0.8 = 0.75.

The basic step of association rule mining is to extract the two sets *A* and *C* and calculate their support and confidence score of rule *R*. Itemset extraction can be done in different ways. One straightforward method is the pattern growth method that starts with a single-item and then keeps expending the set with new items, provided that the frequency of the newly formed set is greater that a support parameter *S*. The algorithm stops when no itemset can be expanded. A recent method, BiMax, was proposed to extract itemsets using biclustering. BiMax takes a dataset represented as binary matrix XY (a value of 1 means item in column Y appears within a row X) and outputs a set of biclusters (*i.e.* sub-matrices).

The BiMax method is similar to the pattern growth in one way: The number of rows in the matrix bicluster actually represents the support value of the itemset that is the elements of the columns of the matrix bicluster. This way, BiMax can provide the most frequent itemset of the dataset and it would be easier to generate rule $R=A \Rightarrow C$, where *A* and *C* are generated from column items in bicluster matrices. After generating the biclusters, output of BiMax was then post-processed to extract frequent item sets and generate the sets*A* and *C* to create association rules. The number of rows of a bicluster readily represents the support value (after being normalized by dividing by the total number of rows in the original matrix).

APPENDIX B

A Mathematical Model for Weighing Association Rules

Let $P(C, T \mid A)$ represent the probability of observing an item set *C* together with a transaction *T*, given that another set of items *A* is observed. Using Bayes' rule and based on the assumption that *C* and *T* are conditionally independent on *A*, this probability can be factored as follows:

$$P(C,T \mid A) \propto P(C \mid A) * P(T \mid A) \quad \text{(B1)}$$

The probability value $P(C|A)$ can be obtained readily from the confidence score of an association rule of the form $A \Rightarrow C$ as described in step one of the method. The value $P(T|A)$ is the probability of observing a transaction *T* given a set of items *A*. The exact identity of sets *T* and *A* is application specific. In this study, we define a transaction *T* as a set of human proteins and the item set *A* as a set of viral proteins. The item sets *T* and *A* were obtained from each of the biclusters extracted from HV and VH matrices representing human-viral and viral human interactions respectively. The itemset *T* represented the rows of the matrix bicluster and *A* represented the columns of the matrix bicluster. The probability $P(T|A)$ does not represent a meaningful quantity by itself if *T* is just a transaction identifier. A more interesting probability model would be a one that represents how much the meta data of a transaction *T* depends on an item set *A*. Let $M(T)$ be a function that retrieves (or synthesizes) meta data about a transaction *T*.

$$M(T) = t_1, t_2, \cdots t_n \quad \text{(B2)}$$

The $M(T)$ function returns a set of tags or features associated with a transaction *T*. In this study, a transaction *T* can be a set of items (*e.g.* human proteins), and $M(T)$ was defined to return the set of gene ontology annotations t_1, t_2,. t_n.Hence, $P(M(T)|A)$ is used instead of $P(T|A)$ to evaluate $P(C, T \mid A)$. In this model, $P(M(T)|A)$ represents the likelihood of observing a set of features of a future transaction given a set of items *A*. Therefore, in this model, a characteristic of a transaction is considered in the process of applying association rules to predict co-occurrence of *T* and *C*.

Then, $P(T|A)$ can be rewritten as:

$$P(T \mid A) \approx P(M(T) \mid A) = P(t_1, t_2, \cdots t_n \mid A) \quad \text{(B3)}$$

Assuming transaction meta data are conditionally independent on item set *A,* the above equation can be written as:

$$P(T \mid A) \approx P(M(T) \mid A) = P(t_1, t_2, \cdots t_n \mid A) = \prod_{i=1}^{n} P(t_i \mid A) \quad \text{(B4)}$$

The probability $P(t_i|A)$ was computed by counting how many times a tag t (gene ontology annotation)co-occurred with a particular set of viral proteins *A* across the dataset, and then the counts were normalized to produce a probability model).

Using the last equation, subsequently,$P(C,T|A) = P(C|A) * P(T|A)$ can be stated as:

$$P(C, M(T) \mid A) \propto P(C \mid A) * P(M(T) \mid A) = P(C \mid A) * \prod_{i=1}^{n} P(t_i \mid A) \quad \text{(B5)}$$

In an association rule mining context, we are including meta information about transactions to weigh the rule. New transactions that contain too few characteristic features that

were observed in a history of transactions are not expected to contain that item set.

The probability value $\prod_{i=1}^{n} P(t_i \mid A)$ poses a problem when a large set of features are observed (synthesized) for a given transaction. This score value would be higher for transactions with fewer features and lower for transactions with numerous features. This is due to arithmetic multiplication operation that involves small fraction values such as those of probability. To compensate for this deficiency, a penalty factor was introduced in the calculation. Assuming that out of *n*-features that are in association with a set *A*, a new transaction T_1 was found to possess *m* features only, where $m < n$. In that case, the probability value is multiplied by a penalty factor. This penalty factor is the probability of the missing feature (a constant value, like $\varepsilon = 0.001$) raised to the power of *n-m*. If *n=m* then the exponent would be zero and hence no penalty incurred. Finally, the overall model is expressed as:

$$P(C, M(T) \mid A) = P(C \mid A) * P(M(T) \mid A) = P(C \mid A) * \prod_{i=1}^{n} P(t_i \mid A) = P(C \mid A) * \prod_{t \in f} P(t \mid A) * (\varepsilon)^{n-|f|} \quad \text{(B6)}$$

Where *n* is the maximum number of features that are observed in a given transaction.

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